Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, SQL, R, Java
Generative AI & NLP: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Google Gemini API, Semantic Search
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Statistical Analysis, Exploratory Data Analysis (EDA), Feature Engineering, Predictive Modeling, Anomaly Detection, Model Evaluation
Machine Learning Techniques: Random Forest, Decision Trees, Naive Bayes, Support Vector Machines (SVM), Time-Series Analysis, BG/NBD, Gamma-Gamma Models
Databases & Data Engineering: SQLite, MySQL, MongoDB, SQL Querying, Data Cleaning, Data Preprocessing, Data Validation, Data Modeling, Data Pipelines
Analytics & Visualization: Power BI, DAX, Tableau, Matplotlib, Seaborn, Business Intelligence, Customer Segmentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.

Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** — 2026